Math Didn't Fail You — Language Did

Presentation Outline

Subtitle: The barrier between "everyday people" and "advanced mathematics" is largely linguistic and psychological, not conceptual.

Tagline: A perspective on math from a girl tired of being the only one excited about it

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PRESENTATION FLOW

Opening: Hook & Reality Check

THE REALITY CHECK

- Ask the room: "Who here thinks they're bad at math?"
- Emphasize the universality of math anxiety
- Frame: This is THE INTRODUCTION to shift their perspective

Slide 1: Title Slide

"Math Didn't Fail You — Language Did."

- Main thesis: The barrier between "everyday people" and "advanced mathematics" is largely linguistic and psychological, not conceptual
- Set tone: personal, frustrated with status quo, evidence-based

Speaker Notes:

- Open with vulnerability: "I'm tired of being the only one excited about math"
- Position this as a perspective shift, not a lecture
- Promise: By the end, you'll see you already 'do' calculus, abstract algebra, and topology in everyday life

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Slide 2: Who here thinks they're bad at math?

Who here thinks they're bad at math?

By the end, you'll see you already 'do' calculus, abstract algebra, and topology in everyday life. I'll prove it with one scary-looking problem that collapses to simple algebra.

THE REALITY CHECK | INTRODUCTION

Speaker Notes:

- This is the hook that engages the audience immediately
- Show of hands: ask who thinks they're bad at math
- Promise a concrete demonstration

- Set expectation: You'll prove the thesis with a real example
- Transition: "Let's start by understanding what's really happening"

Slide 3: What Really Drives Performance?

64% report math anxiety
But anxiety alone is not the direct performance killer. The stronger mechanism is **math engagement**.

KEY QUOTE (Dark Box):

> "People with high math anxiety often know how to do the math — it's an emotional reaction that interferes, not a lack of skill."
> — American Psychological Association

MEDIATING MECHANISM (SPRINGER, 2025):

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Math Anxiety → Math Engagement → Math Performance
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Anxiety still matters because it suppresses engagement. If we can standardize high engagement, anxiety's performance impact drops substantially.

ARGUMENT BOX:

> "Math anxiety" is a major indirect force, not always a direct one. Engagement is the lever we can design and control.

Supporting Points:

Direct Effect Is Weaker

- Students with anxiety do not automatically perform poorly in every condition
- Performance is more strongly tied to whether they stay cognitively engaged with the task

Engagement Mediates

- The 2025 study shows anxiety predicts lower engagement, and lower engagement predicts lower performance
- The pathway is primarily mediated through engagement

Source Citation:
"Influence of mathematics anxiety on mathematics performance: mediating effects of mathematical engagement" (Mathematics Education Research Journal, 2025). DOI: 10.1007/s13394-025-00536-1

Speaker Notes:

- Don't dismiss anxiety — it's real
- But point out: the problem isn't that anxious people CAN'T do math
- The problem is they DISENGAGE
- This means: we can intervene by redesigning HOW we present math

- If we keep people engaged, anxiety's impact drops dramatically

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Slide 4: The Core Problem: It's a Language Barrier

Four Authoritative Quotes:

Sheila Tobias (Author of *Overcoming Math Anxiety*, 1978)

> "The barrier to math is often linguistic, not conceptual. Many students can handle the mathematical concepts but become unnerved by the unfamiliarity of the symbols, terminology, and structure."

Anna Sfard (*Thinking as Communicating*, 2008)

> "Thinking is a form of communication. Students 'hit a wall' not because they can't do the math, but because they can't parse the formal language it's written in."

American Psychological Association (*Preventing Math Anxiety*, 2023)

> "Math anxiety is an emotional problem, not a cognitive one. When anxiety is reduced, anxious individuals perform just as well as their non-anxious peers."

Ferreira, R. A., Rodriguez, C., Guzmán, B., Sepúlveda, F., & Peake, C. (*The Interplay of Vocabulary, Working Memory, and Math Anxiety*, 2025)

> "Jargon is the gatekeeper, not intelligence. The more specialized the vocabulary, the more it creates an 'insider vs. outsider' dynamic that can intimidate and alienate learners."

KEY TAKEAWAY (Dark Box - Right Side):

> Most people who say "I'm bad at math" are actually saying "math language scares me."

The ability is already there. The interface is the problem.

Speaker Notes:

- This is the thesis slide
- Hammer home: it's NOT about intelligence
- It's about ACCESS — specifically linguistic access
- Set up the examples coming next

Slide 5: Scary Terms You Already Understand

Nine Examples in 3x3 Grid:

Derivative	Integral	Eigenvalue / Eigenvector

Derivative	Integral	Eigenvalue / Eigenvector
YOU KNOW THIS AS: **Your Speedometer.** It shows the rate of change of your position at this exact instant. How fast are you going *right now?* That's a derivative.	YOU KNOW THIS AS: **Accumulation.** Estimating the total cost of gas for a trip by adding up small chunks along the way. Summing up tiny pieces to get a total.	🔀 YOU KNOW THIS AS: **Stretching a Rubber Band.** The direction along the band that only *stretches* — never rotates — is the eigenvector. How much it stretches is the eigenvalue.

Combinatorics	Ergodicity	Graph Theory
👕 YOU KNOW THIS AS: **Outfit Planning.** "4 shirts and 3 pants = 12 outfits." Counting arrangements and possibilities is the heart of combinatorics.	🚌 YOU KNOW THIS AS: **Shuffling Cards.** Shuffle a deck enough times and you eventually experience every typical outcome — the average over time equals the average across all possibilities at once. You live inside ergodic processes daily.	🔗 YOU KNOW THIS AS: **Social Networks.** "Friend of a friend." Finding the shortest connection path between people. Dots (people) connected by lines (relationships).

Fourier Transform	Topology	Diffeomorphism
🎵 YOU KNOW THIS AS: **Hearing Music.** Your ear breaks a complex sound wave (a chord) into individual notes. That decomposition is a biological Fourier transform.	🍩 YOU KNOW THIS AS: **Shape Equivalence.** A donut and a coffee mug are "the same" because each has one hole. You could mold one into the other without tearing or gluing.	🍕 YOU KNOW THIS AS: **Stretching Pizza Dough.** Reshaping something smoothly without tearing or cutting, and being able to smoosh it back. A reversible transformation of shape is a diffeomorphism.

Speaker Notes:

- Move through these quickly but confidently
- Emphasize: "You ALREADY understand the concept"
- The only difference is the label
- This builds momentum and confidence

Slide 6: Research Evidence

RESEARCH EVIDENCE

The barrier isn't ability. It's how math is presented.

Two independent research reviews reach the same conclusion: symbolic format and notation density—not the underlying ideas—are the triggers for avoidance and disengagement.

Two Studies (Side by Side):

FRONTIERS IN PSYCHOLOGY — 2022
Trait anxiety is activated by format, not concept

Trait math anxiety amplifies performance failures on complex tasks—but the trigger is symbolic presentation and formal notation, not the mathematical ideas themselves.

JRSMTE LITERATURE REVIEW — 2022
Notation density directly raises cognitive load

When unfamiliar symbols don't map to a student's intuitive model, working memory fills up translating notation—leaving little capacity for actual mathematical reasoning.

Speaker Notes:

- This is your credibility slide
- You're not just opining — published research backs this up
- The problem is PRESENTATION, not ability
- Set up the solution section

Slide 7: Engagement is the Mediating Variable

RESEARCH SYNTHESIS

Engagement is the mediating variable. Language controls engagement.

The strongest predictor of whether students persist in math is not ability—it's engagement quality. And engagement quality is directly shaped by language, format, and the order in which concepts are introduced. Pedagogy can break the anxiety-performance chain.

SUPPORTING EVIDENCE

FRONTIERS IN PSYCHOLOGY — 2022
Format as a trigger condition

The state vs. trait anxiety distinction reveals that symbolic presentation acts as the situational trigger for chronic fear responses—not concept difficulty.

SPRINGER MERJ — 2025
Engagement mediates anxiety → performance

When language-first pedagogy sustains engagement, anxiety's negative effect on performance is significantly buffered. The link can be broken.

META-ANALYSIS (ERIC) — 2023
Language interventions are repeatable

Vocabulary support and gradual notation introduction consistently improve outcomes across different populations, settings, and math subjects.

🔗 SPRINGER MERJ, 2025

Speaker Notes:

- This is the "so what?" moment
- We can INTERVENE
- If we change how we present math, we keep people engaged
- And engagement breaks the anxiety chain

- Transition: "So what do we actually DO about this?"

Slide 8: Math You Already Use Daily (Without Realizing It)

Nine Examples in 3x3 Grid:

Quick Estimation	Spatial Geometry	Symbolic Modeling
📊 YOU DO THIS WHEN: **Eyeballing a tip or travel time.** You deliberately accept a small error in exchange for a fast, useful answer — the exact trade-off that defines Approximation Theory.	🚘 YOU DO THIS WHEN: **Parallel parking or packing a suitcase.** Judging angles, distances, and fit on flat surfaces is Euclidean Geometry — every measurement you make in the physical world relies on it.	🗺️ YOU DO THIS WHEN: **Reading a map or a traffic sign.** Substituting a simpler stand-in that preserves the key structure of a complex situation is the core move of Representation Theory.

Pattern Recognition	Budget Arithmetic	Inevitable Patterns
🎶 YOU DO THIS WHEN: **Predicting the next song or guessing a plot twist.** Any process whose next step has predictable probabilities rather than a fixed outcome is a Stochastic Process — you navigate constant uncertainty constantly.	💰 YOU DO THIS WHEN: **Comparing unit prices at the grocery store.** Finding the best option within a spending constraint is Linear Optimization — the same math behind airline pricing and logistics networks.	👥 YOU DO THIS WHEN: **Noticing that in every large group, certain dynamics emerge.** Ramsey Theory proves that complete disorder is impossible at scale — order always appears, no matter how the pieces are arranged.

Cascade Effects	Clock Arithmetic	Curvature Intuition
⚡ YOU DO THIS WHEN: **Leaving two minutes late and having a completely different day.** Systems sensitive to tiny initial differences are the subject of Chaos Theory — deterministic, yet practically unpredictable.	🕐 YOU DO THIS WHEN: **Calculating time zones or scheduling across a 12-hour cycle.** "10am + 5 hours = 3pm" — numbers that wrap around is Number Theory's modular arithmetic in everyday form.	🌍 YOU DO THIS WHEN: **Knowing that a flight path curves north on a flat map.** The shortest route on a sphere is a geodesic — the central object of Differential Geometry, solved by your GPS every trip.

Speaker Notes:

- This is the empowerment moment
- Don't just list — DEMONSTRATE
- Ask: "How many of you have done [action]?" for a few examples
- Make them realize: they're ALREADY mathematical thinkers
- The only difference is vocabulary

Slide 9: How We Fix The Interface

Four evidence-backed moves any teacher can make.

Four Strategies (2x2 Grid):

💬 Language First, Notation Second

Introduce concepts in plain English before introducing symbols. Build the intuition, then label it mathematically.

📖 Define Vocabulary Explicitly

Don't assume it. Decode the jargon immediately so students can focus on the logic, not on deciphering unfamiliar words.

⚓ Use Familiar Anchors

Connect abstract rules to concrete reality — like using a speedometer to explain what a derivative actually means.

💚 Normalize the Anxiety

Acknowledge that fear is an emotional block, not an ability block. It's okay to feel it — it doesn't mean you can't do it.

FOOTER QUOTE:

> "It is time to stop gatekeeping intelligence behind jargon. Mathematical thinking is not a rare talent — it is a human universal."

Speaker Notes:

- This is the ACTION slide
- These are tangible, research-backed strategies
- Neither you nor the audience needs a PhD to implement these
- Emphasize: these aren't "nice to have" — they're PROVEN to work

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Slide 10: You're Not Bad at Math

You're not bad at math.

You already think mathematically every day. Learn the dialect, and the field opens up.

STOP SAYING
~~"I'm bad at math."~~

↓

START SAYING
"I'm learning the language."

● THANK YOU ●

Speaker Notes:

- This is the emotional landing

- Full circle back to the opening
- Reframe: it's not about ability, it's about ACCESS
- Leave them empowered, not lectured
- End on the call to action: change how you talk about math, to yourself and others

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DELIVERY NOTES

Timing (10-15 minute version)

- **Slides 1-2:** 2 minutes (title + hook)
- **Slides 3-4:** 3-4 minutes (thesis + problem statement)
- **Slides 5-6:** 3-4 minutes (examples + evidence)
- **Slides 7-8:** 3-4 minutes (mechanisms + daily math)
- **Slides 9-10:** 3-4 minutes (solutions + closing)

Tone

- **Personal, not preachy:** You're sharing a frustration and a revelation
- **Evidence-backed, not opinion:** Every claim is sourced
- **Empowering, not condescending:** The audience already has the ability

Key Rhetorical Moves

1. **Open with vulnerability:** "I'm tired of being the only one excited"
2. **Use "you already do this" framing** throughout
3. **Cite concrete research** to build credibility
4. **End with linguistic reframe:** "I'm learning the language"

RESEARCH SOURCES

Statistics on Math Anxiety

- 64% of Americans experience math anxiety (13% severe)
- Women: 70% vs. Men: 57%
- 47% say it held them back academically/professionally
- 37% say it hinders financial decision-making

Sources:

- Prodigy Game
- District Administration

Key Academic Sources

Sheila Tobias - *Overcoming Math Anxiety* (1978)

- Foundational work establishing math anxiety as a linguistic barrier

Anna Sfard - *Thinking as Communicating* (2008)

- Mathematics as discourse/language

APA - *Preventing Math Anxiety* (2023)

- Math anxiety is emotional, not cognitive
- Link

Ferreira et al. - *The Interplay of Vocabulary, Working Memory, and Math Anxiety* (2025)

- Math-specific vocabulary as gatekeeper
- Link

Frontiers in Psychology - *Unraveling the Role of Math Anxiety* (2022)

- State vs. trait anxiety; presentation as trigger
- Link

Springer MERJ - *Influence of Mathematics Anxiety on Performance* (2025)

- Engagement as mediating variable
- Language-first pedagogy buffers anxiety effects
- Link

ERIC Meta-Analysis (2023)

- Language interventions consistently effective
- Link

APPENDIX: Extended Examples (For Q&A or Longer Format)

Additional "Scary Terms You Already Understand"

Heuristic

- Practical rule of thumb
- "Don't grocery shop hungry" / "If it sounds too good to be true, it probably is"

Asymptote

- A line you approach but never reach
- "Almost there but never quite" — like halving distance to a wall forever

Universal Quantifier (∀)

- Symbol meaning "for all" / "every"
- "Every restaurant here is expensive" — you use this in speech constantly

Ansatz

- An educated starting guess
- Estimating a tip, guessing drive time, eyeballing if furniture fits

Genus

- Number of holes in a shape
- Sphere = 0 holes, donut = 1 hole, pretzel = 2 holes

Manifold

- Looks flat up close, curves globally
- You live on one: Earth appears flat locally but is a sphere globally

Orthonormal

- Perpendicular directions at right angles, evenly scaled
- Every map grid, every 3D graph axis since middle school

Cardinality of the Continuum

- Different sizes of infinity
- Infinitely many whole numbers vs. infinitely many numbers between 0 and 1

Galois Theory

- Symmetries among equation solutions
- Like kaleidoscope patterns — rotation preserves structure

Real Analysis (Epsilon-Delta)

- "You can get as close as you want to a target"
- Thermostat: no matter how precise your comfort demand, you can dial it in

Ramsey Theory

- Order inevitably appears in large enough structures
- In any 6 strangers, guaranteed 3 mutual friends OR 3 mutual strangers

Chaos Theory

- Deterministic but sensitive to initial conditions
- Leaving 2 minutes late cascades into completely different day

Ring Theory

- Algebraic structures with addition and multiplication
- Every arithmetic operation on integers is ring theory

Module Theory

- Vector spaces where scalars come from rings
- Budget spreadsheet with different multipliers per category

CLOSING THOUGHT

The goal of this presentation is not to teach math — it's to **change the conversation about who can do math**.

When we shift from "I'm bad at math" to "I'm learning the language," we remove the stigma and open the door.

Math didn't fail you. Language did.

And language can be learned.